

5.6 CYLINDRICAL COORDINATES

Use the following to write the integrand, limits, and element of area in terms of cylindrical polar coordinates (r, θ, z) :

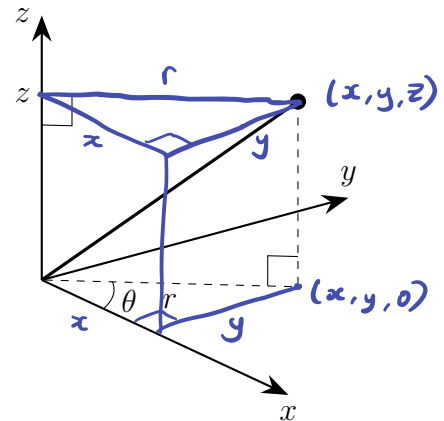
$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

where



$$0 \leq r, \quad 0 \leq \theta < 2\pi \text{ or } -\pi < \theta \leq \pi, \quad -\infty < z < \infty.$$

$$dV = r dr d\theta dz$$



r is the distance from the z axis:

$$r^2 = x^2 + y^2$$

Example 5.6.1

Find the centre of mass of the cone of constant density ρ , radius 1 and height 2, oriented

as follows: "average position" weighted by density

The cone mass is rotationally symmetric about the z axis

$$\Rightarrow x_{\text{com}} = 0, \quad y_{\text{com}} = 0$$

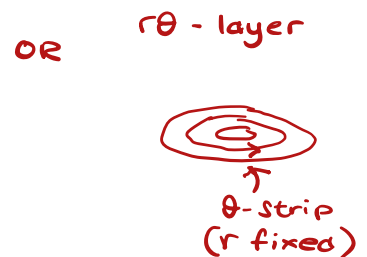
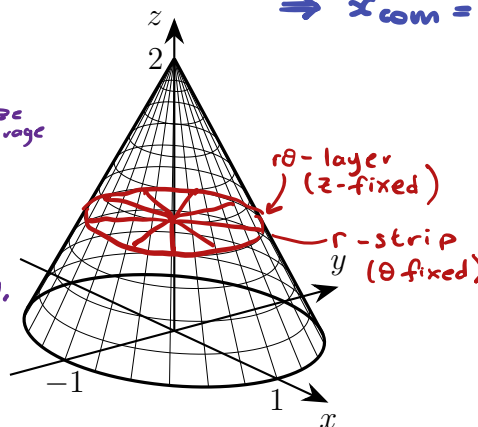
In general:

$$z_{\text{com}} = \frac{\iiint_R \rho(x, y, z) z dV}{M}$$

M comes from an integral

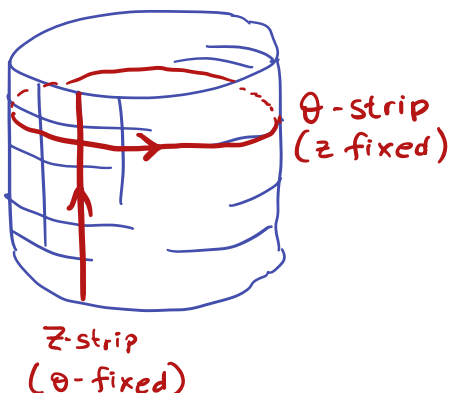
we have to normalize in order to get average z position

When we are doing layers, we hold one of our variables fixed.

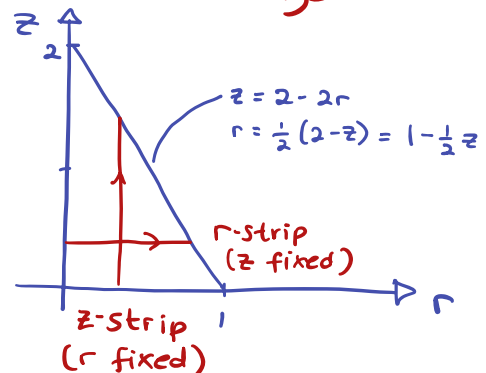


OTHER ALTERNATIVES:

r fixed $\Rightarrow \theta z$ -layer



θ fixed $\Rightarrow rz$ -layer



Solution.

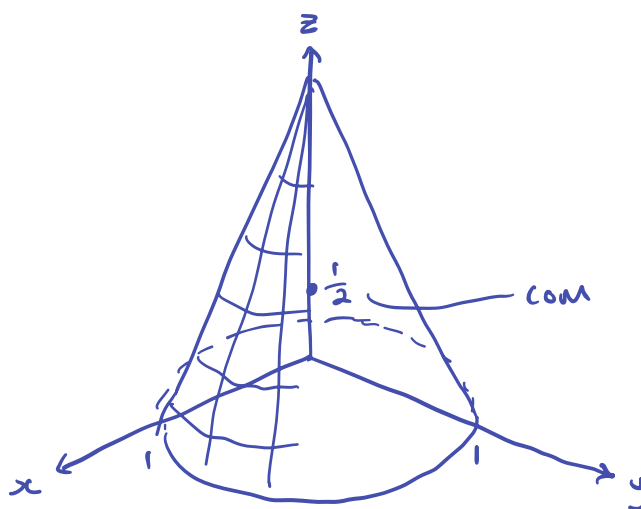
r - layers (z -fixed), r -strips (fixed θdz)
total mass

$$\begin{aligned}
 M &= \iiint_R \rho(x,y,z) dV \\
 &= \int_{z=0}^2 \int_{\theta=0}^{2\pi} \int_{r=0}^{1-\frac{1}{2}z} \rho \overbrace{r dr d\theta dz}^{dV} \\
 &= \rho \int_{\theta=0}^{2\pi} 1 d\theta \int_{z=0}^2 \int_{r=0}^{1-\frac{1}{2}z} r dr dz \\
 &= \\
 &= \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \text{make sure you can do these steps} \\
 &= \\
 &= \frac{2}{3}\pi\rho
 \end{aligned}$$

z -coordinate of centre of mass

$$\begin{aligned}
 z_{\text{com}} &= \frac{1}{M} \iiint_R \rho(x,y,z) z dV \\
 &= \frac{1}{\frac{2}{3}\pi\rho} \int_{z=0}^2 \int_{\theta=0}^{2\pi} \int_{r=0}^{1-\frac{1}{2}z} \rho z r dr d\theta dz \\
 &= \frac{3\rho}{2\pi\rho} \int_{\theta=0}^{2\pi} 1 d\theta \int_{z=0}^2 z \int_{r=0}^{1-\frac{1}{2}z} r dr dz \\
 &= \\
 &= \\
 &= \\
 &= \\
 &= \\
 &= \\
 &= \\
 &= \frac{1}{2}
 \end{aligned}$$

Constant density shortcut:
 $M = \rho V$ $V = \frac{1}{3}\pi r^2 h$ (cone)
 $= \frac{1}{3}\pi (1)^2 (2)$
 $= \frac{2}{3}\pi$



constant density $\Rightarrow$ com = centroid $\blacklozenge$

5.7 SPHERICAL POLAR COORDINATES

useful for when you have a sphere / part of a sphere, bad for everything else

Use the following to write the integrand, limits, and element of area in terms of spherical polar coordinates (r, θ, ϕ) , where r is the radial length, θ is the azimuthal angle in the xy -plane, and ϕ is the inclination or zenith angle (the angle the radius line makes with the positive z -axis). Think about what surfaces of constant r , θ , and ϕ look like.

phi
 ϕ or φ

r is distance from origin:

$$r^2 = x^2 + y^2 + z^2$$

$$x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta, \quad z = r \cos \phi$$

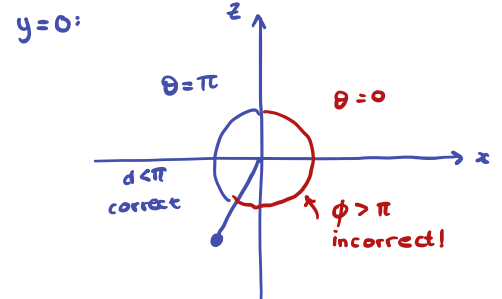
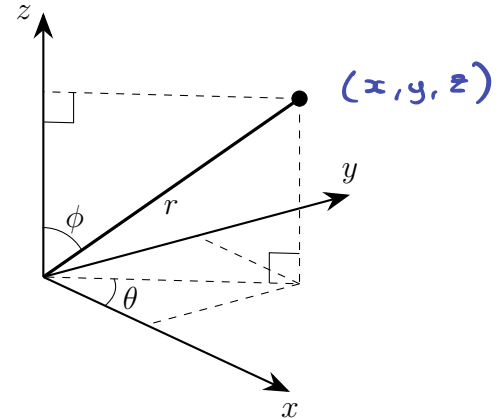
where

$$0 \leq r < \infty, \quad 0 \leq \theta < 2\pi \text{ or } -\pi < \theta \leq \pi, \quad 0 \leq \phi \leq \pi$$

OR $-\pi \leq \theta \leq \pi$

$$dV = r^2 \sin \phi \, dr \, d\theta \, d\phi.$$

catches people

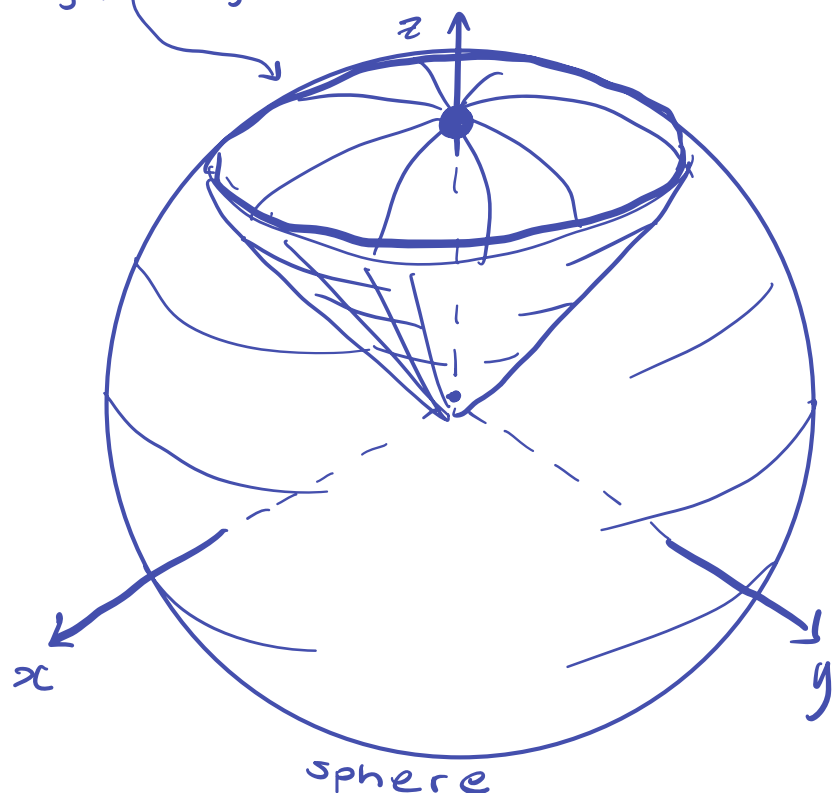
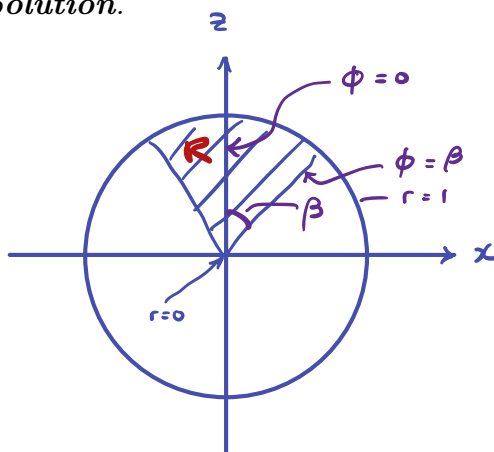


Example 5.7.1

Find the volume and the centroid of the sector of a sphere of constant density 1, radius 1, and semi-angle β . centre at origin: $x^2 + y^2 + z^2 = 1 \Rightarrow r^2 = 1 \Rightarrow r = 1$

Solution.

when $y=0$:



$$\begin{aligned}
 V &= \iiint_{\mathcal{R}} 1 dV \\
 &= \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\beta} \int_{r=0}^1 r^2 \sin \phi dr d\phi d\theta \\
 &= \int_{\theta=0}^{2\pi} 1 d\theta \int_{\phi=0}^{\beta} \sin \phi d\phi \int_{r=0}^1 r^2 dr \\
 &= \\
 &= \\
 &= \frac{2\pi}{3} (1 - \cos \beta)
 \end{aligned}$$

$$z_{cen} = \frac{1}{V} \iiint_{\mathcal{R}} z dV$$

Again, symmetry means $x_{cen} = 0$ and $y_{cen} = 0$.

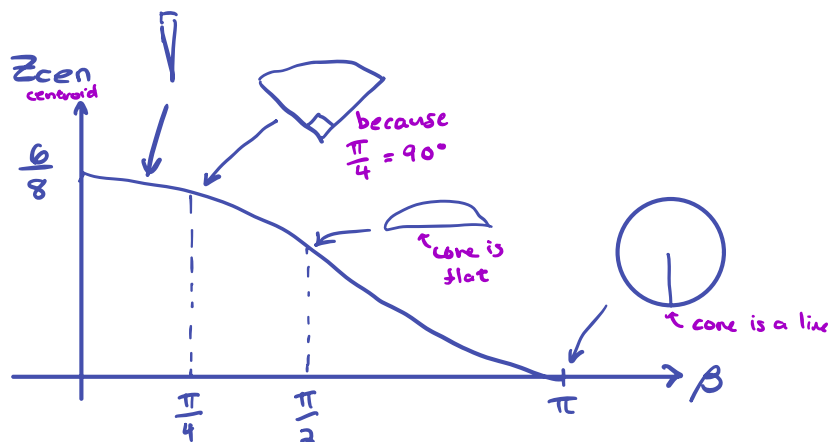
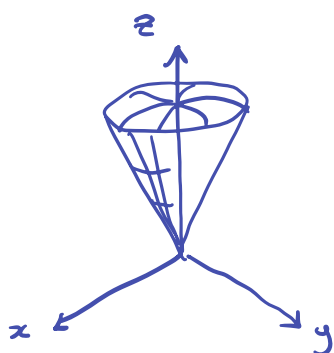
$$\begin{aligned}
 z_{cen} &= \frac{1}{V} \iiint_{\mathcal{R}} z dV \\
 &= \frac{1}{V} \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\beta} \int_{r=0}^1 \overbrace{r \cos \phi}^z \overbrace{r^2 \sin \phi dr d\phi d\theta}^{dV} \\
 &= \frac{1}{V} \int_{\theta=0}^{2\pi} 1 d\theta \int_{\phi=0}^{\beta} \frac{1}{2} \sin 2\phi d\phi \int_{r=0}^1 r^3 dr
 \end{aligned}$$

TRIG IDENTITY
FROM FORMULA SHEET
 $\sin \phi \cos \phi = \frac{1}{2} \sin 2\phi$

=
should be able to do this step,
if you have trouble, go to the drop ins
@ JF 443 MON-FRI 4-6PM

-Michael

$$= \frac{3}{8} (1 + \cos \beta)$$



UNIVERSITY OF CANTERBURY

EMTH210

ENGINEERING MATHEMATICS

Topic 6: Laplace transforms

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6.1 LAPLACE TRANSFORM - important when dealing with things in electrical

The most important use of the *Laplace Transform* is in solving differential equations. The Laplace Transform, or LT, turns the problem of solving an ODE into one of solving an algebraic equation, then looking things up in a table.

The Laplace Transform turns a formula in terms of t into a formula in terms of s , and the inverse Laplace Transform does the opposite: , so transform, rearrange, then transform back

S is something else (not really physically intuitive) fancy L: $\mathcal{L}$

$$F(s) = \mathcal{L}\{f(t)\} \quad \text{and} \quad f(t) = \mathcal{L}^{-1}\{F(s)\}$$

transform of a function, to give us another function a transformation not a function t is physically meaningful (usually time)

We will see the definition and theory of the Laplace Transform soon, but first we will look at how it is used in practice.

One powerful new capability Laplace Transforms give us is the ability to solve differential equations involving piecewise functions, such as terms that get switched on or off at particular times. This happens frequently in real-world applications when power to a system or part of a system is switched on or off, or when a valve is opened or closed, or when a physical connection is made or broken.

The basic idea behind our use of LTs to solve ODEs is that we transform the ODE (which has an unknown function of t and its derivatives) into an algebraic equation (in terms of an unknown function of s), rearrange it a bit, and then use the inverse Laplace transform to turn the algebraic equation (in s) into an algebraic equation (in t) — the solution to the ODE. Moreover, the method automatically inserts initial conditions.

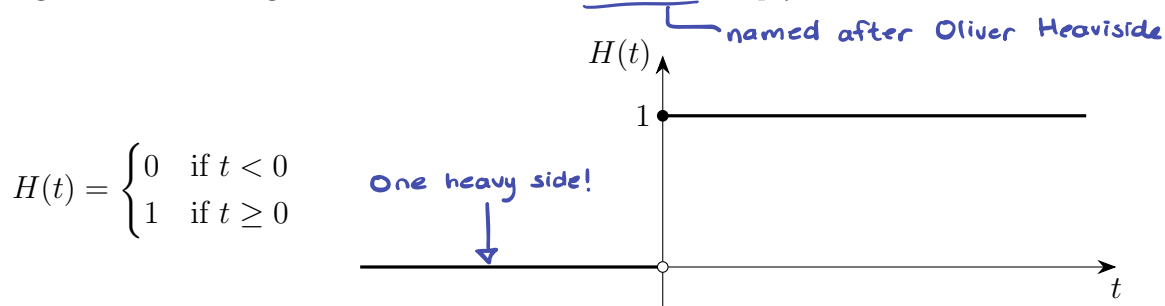
$$y(t) \rightarrow Y(s)$$

$$Y(s) \rightarrow y(t)$$

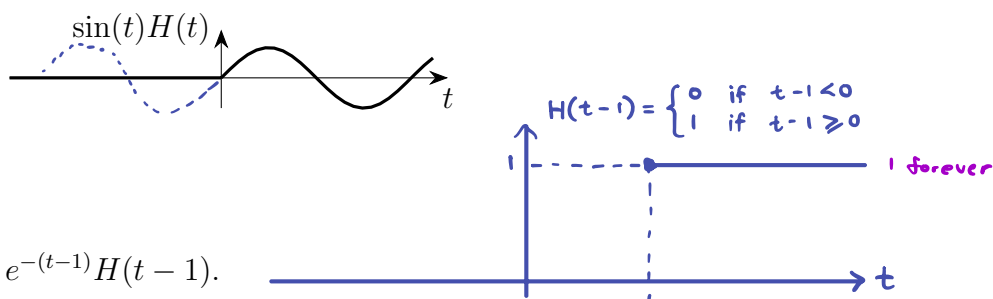
via table

6.2 HEAVISIDE STEP FUNCTION

To work with piecewise functions using Laplace Transforms, we need to write them as single formulas using a function called the *Heaviside step function*¹:



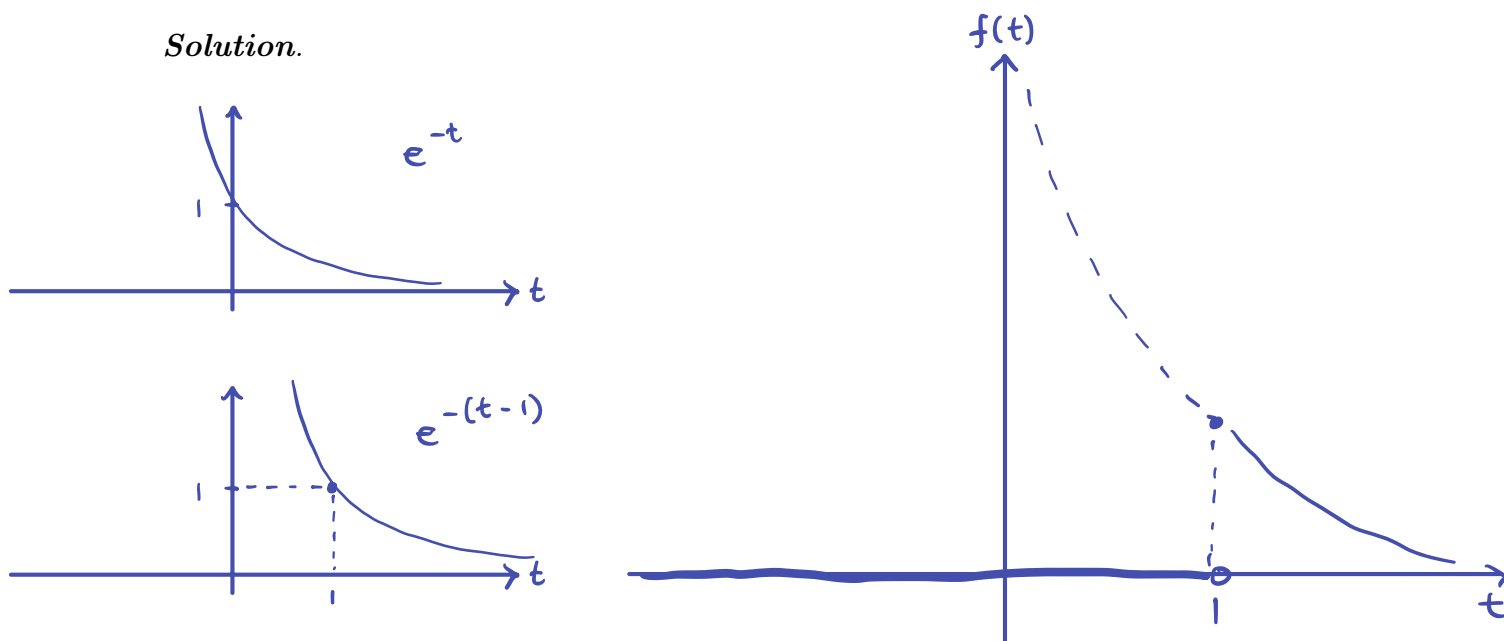
This function acts like a switch: anything we multiply by it gets multiplied by either 0 or 1, depending on where it is evaluated. E.g. $\sin(t)H(t)$ looks like this:



Example 6.2.1

Sketch the graph of $f(t) = e^{-(t-1)}H(t-1)$.

Solution.



¹Sometimes called the *unit* step function, $u(t)$ or $U(t)$.